Extracting New Product Announcements from SEC Form 8-K Filings with LLMs

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GitHub: https://github.com/fersobe12-create/sec-8k-product-extractor

1. Objective and output

I built a Python pipeline (extract_8k_products.py) that reads recent Form 8-K filings of S&P 500 companies, finds new product announcements, and saves them as a CSV with five fields: company_name, stock_name, filing_time, new_product and product_description (under 180 characters). It combines BeautifulSoup parsing, spaCy Named Entity Recognition (NER) and a local LLM (LLaMA 3.1 8B served by Ollama), so it runs at no cost on a laptop.

2. Data collection

- **Ticker to CIK.** https://www.sec.gov/files/company_tickers.json maps each ticker to its CIK and official company name (e.g. AAPL → 0000320193). Company Name and Stock Name come from this file, not from the LLM, so they are always exact.
- **8-K list.** The EDGAR Atom feed (browse-edgar?action=getcompany&CIK={cik}&type=8-K&count={count}&output=atom) is parsed with BeautifulSoup for each filing's accession number, <updated> timestamp (used as Filing Time), index-page link and reported items.
- **Fair access.** Every request sends a User-Agent with my e-mail and waits 0.15 s (under 10 requests/second). Responses are cached on disk so reruns don't hit the SEC again.

3. Preprocessing

Most 8-Ks report earnings, executive changes or votes, so the text passes three filters before any LLM call:

1. **Item filter:** keep filings reporting Item 7.01 (Reg FD), 8.01 (Other Events) or 9.01 (Exhibits); skip filings that only report items such as 5.02 or 5.07.
2. **Document selection:** from the filing index, download the 8-K body and EX-99.x exhibits; press releases (EX-99.1) are read first.
3. **Cleaning and paragraph filter:** BeautifulSoup strips scripts, styles and hidden XBRL headers. Only paragraphs with launch language (launch, introduce, unveil, now available, FDA approval, next-generation…) and the paragraph after each are kept. Forward-looking-statement boilerplate is removed, and the text is capped at 6,000 characters.

4. NER and LLM extraction

NER. spaCy en_core_web_sm tags ORG and PRODUCT entities in the filtered text. In the example "Apple Inc. has announced the launch of its latest MacBook Pro," it labels Apple Inc. as an organization. Because the small model often misses product names (drugs, chips, software), the entities are passed to the LLM as hints rather than used alone.

LLM. Each filing is one request to Ollama's /api/chat endpoint with temperature 0 and JSON-only output. The prompt has a system message defining a new product (the filing company's own named good, service, drug, device, software or platform being launched or approved) and listing exclusions (earnings, acquisitions, plans, partnerships, disclosure channels, products mentioned in passing). It also includes one worked example (the MacBook Pro sentence) and the company name, NER hints and filtered text. The model returns {"products": [{"name", "description"}]}, or an empty list when there is no product.

5. Validation and clean-up

- **Grounding:** the product name must appear in the filing text, otherwise it is dropped as a likely hallucination. Generic names and the company's own name are rejected, and descriptions are cut at a word boundary to under 180 characters.

- **Clean-up across filings (added after reviewing the first run):** drop share plans, partnerships, sponsorships and disclosure channels such as YouTube; keep a product only at the first filing that mentions it; and of two overlapping names ("Similac" vs "Similac 360 Total Care") keep the more specific one.
- Outputs: new_products.csv (final), new_products_raw.csv (before clean-up), a |-separated table, and processing_log.csv with each filing's status. Offline unit tests cover parsing, filtering, validation and clean-up.

6. Results

The submitted run took the S&P 500 in index order, the 10 most recent 8-Ks per company, and stopped after 100 analysed filings (12 companies, about 4 minutes on an Apple-silicon MacBook Air).

Stage	Count
8-K filings listed	117
Skipped by item filter	17
Analysed	100
No launch language (no LLM call)	72
Sent to the LLM	28
Filings with ≥ 1 product	8
Product rows before / after clean-up	22 / 13

The keyword pre-filter avoided 72% of LLM calls, which made a local model practical. I checked every row by hand against its press release. Of the 22 raw rows, 4 were not the company's own products (YouTube, Accenture's share incentive plan, 3M's Formula 1 sponsorship and the Airbus A220 that 3M supplies insulation for), and 6 repeated a product already captured. The final CSV has **13 products, 12 of them correct (92% precision)**, for example Abbott's Libre Duo glucose-ketone sensor, Amulet 360 occluder and SimpleScreen CRC test, Aflac's Anshin Palette and Miraito insurance products, and 3M's PELTOR ComTac VI headsets. The remaining error is the A220 row.

7. Limitations and improvements

The pipeline favours precision, so launches described in unusual wording can be missed. The model can confuse a supplier's part with the customer's product (the A220 case), which the updated prompt now addresses. The small spaCy model is weak on PRODUCT entities, and a transformer NER model would give better hints. Sampling 2–3 filings per company across all sectors, instead of 10 filings from the first 12 companies alphabetically, would give a broader set of products.